

CerbaSeal

Pricing and Commercialization Source Data

Evidence package for rebuilding the commercialization, pilot readiness, and pricing analysis document. All facts extracted from repository artifacts and verified commands. No estimates. No invented numbers. No loose summaries.

Generated	2026-06-08
Product	CerbaSeal Core v0.1.0
Commit	485cde3faaa0524ffd5f5b6132673789f89adab9
Branch	main
Tests	432 passed / 432 (17 files)
Audit checks	16 / 16 passed
Invariants	12 / 12 covered
stableChecksum	e5aca8b2ad5f7abb528322be754d06463cb01367e038daf1472f925206c64e2c

Package Identity

Field	Value	Source
Package name (internal)	cerbaseal-review	package.json !"name"
Product name	CerbaSeal Core	package.json !"description"
Version	0.1.0	package.json !"version"
Description	CerbaSeal review repo - Drop 01 execution enforcement spine	package.json
Author / owner	Jesse Lamont / Lamont Labs	README, partner kit docs
Module type	ESM ("type": "module")	package.json

Section 4Deployment Readiness

Version Control State

Section 5Client Onboarding Process

Section 6Founder Dependencv Analysis

Field	Value	Command
Current commit (full)	485cde3faaa0524ffd5f5b6132673789f89adab9	git rev-parse HEAD
Current commit (short)	485cde3	git log --oneline -1
Branch	main	git branch --show-current
Last commit message	"Published your App"	git log --oneline -1

Section 11External Market Research Placeholders

Runtime Environment

Section 12Known Limitations and Roadmap

Field	Value	Command
Node.js	v22.22.0	node --version
pnpm	10.26.1	pnpm --version

Purpose: Evidence package for rebuilding the CerbaSeal commercialization, pilot readiness, and pricing analysis document with maximum accuracy. All facts are extracted directly from repository artifacts and verified commands. No estimates, no invented numbers, no loose summaries.

Field	Value	Command
Tests passed	432	pnpm test
Total tests	432	pnpm test
Test files	17	pnpm test
Duration	~5.2 seconds	pnpm test
Status	PASS (all)	pnpm test

Full test suite output (tail):

Test Files	17 passed (17)
Tests	432 passed (432)
Duration	5.22s (transform 1.19s, setup 0ms, collect 2.75s, tests 412ms, environment 4ms, prepare 4.14s)

Repository Audit

Field	Value	Command
Audit checks passed	16	pnpm audit:repo
Total checks	16	pnpm audit:repo
Status	PASS	pnpm audit:repo

Full audit output:


```

' PASS 1. Full test suite passes – 432 tests passed
' PASS 2. TypeScript compiles without errors – tsc --noEmit clean
' PASS 3. README anchor strings present – all 4 anchors found
' PASS 4. All portal routes respond 200 – 9 routes checked
' PASS 5. No src/ file unreferenced in tests or examples – all 29 src files referenced
' PASS 6. Invariant registry exists and is non-empty – 12 invariants registered
' PASS 7. Known-limitations section present in README – section found
' PASS 8. Test count in README matches actual – both report 432 tests
' PASS 9. demo:web:validate passes – all assertions passed
' PASS 10. demo:support:validate passes – 13 assertions passed
' PASS 11. review:validate passes – 110 assertions passed
' PASS 12. No architectural import boundary violations – all boundaries clean
' PASS 13. All invariants linked to covering tests – 12 / 12 invariants covered
' PASS 14. No stale test-count references in documentation – all references match actual count
' PASS 15. Pilot documents exist and contain required sections – 3 pilot docs verified
' PASS 16. cerbaseal.policy.json parses without error – policy file valid

```

Proof Snapshot and Stable Checksum

Command: `pnpm export:proof` (writes to `docs/reports/proof-snapshot.json`)

Field	Value
stableChecksum (enforcement state only)	e5aca8b2ad5f7abb528322be754d06463cb01367e038daf1472f925206c64e2c
manifestChecksum (full body)	53f5733305c6e585b21ed8c4239a1197fd48efb91da690044dc4f116def20cae
Proof file path	docs/reports/proof-snapshot.json
Proof status	VERIFIED (checksums match)

Verify command: `pnpm verify:proof`

```

demo:web:validate: 106 assertions (pass)
demo:support:validate: 13 assertions (pass)
review:validate: 110 assertions (pass)
Status: VERIFIED – Both checksums match. The snapshot has not been tampered with.

```

Proof snapshot metadata (from `docs/reports/proof-snapshot.json`):

```

{
  "generatedAt": "2026-06-08T20:48:34.055Z",
  "gitCommit": "485cde3faaa0524ffd5f5b6132673789f89adab9",
  "gitBranch": "main",
  "version": "0.1.0",
  "testSuite": { "passed": 432, "total": 432, "files": 17 },
  "auditChecks": { "passed": 16, "total": 16 },
  "invariants": { "total": 12, "allCovered": true },
  "validators": {
    "demo:web:validate": { "passed": true, "assertions": 106 },
    "demo:support:validate": { "passed": true, "assertions": 13 },
    "review:validate": { "passed": true, "assertions": 110 }
  },
  "manifestChecksum": "53f5733305c6e585b21ed8c4239a1197fd48efb91da690044dc4f116def20cae",
  "stableChecksum": "e5aca8b2ad5f7abb528322be754d06463cb01367e038daf1472f925206c64e2c"
}

```


2.1 Enforcement Gate — Implemented Code

What it is: The core runtime enforcement function. Accepts a GovernedRequest, evaluates it synchronously against 12 invariants and policy configuration, and returns a deterministic GateResult with finalState (ALLOW / HOLD / REJECT), reason codes, decision envelope, and evidence bundle.

Key property: The same input always produces the same output. evaluate() is a pure synchronous function — no external I/O, no side effects on the decision itself.

node -version

v22.22.0

pnpm --version

10.26.1

Item	Detail
Primary source file	src/services/execution/
Config source	src/config/cerbaseal-config.ts
Policy source	src/config/cerbaseal-policy.ts
Domain types	src/domain/types (not a single flat file — see builders, constants, errors, formatters subdirectories)
Test file	test/execution-gate-service.test.ts (19 tests)
Adversarial tests	test/adversarial-integrity.test.ts
Security tests	test/security/ (4 files: contextual-boundary, fail-closed, misuse-scenarios, non-forgery)
Integration tests	test/integration/full-flow.test.ts, test/integration/system-integration.test.ts
Status	Implemented, verified, production-grade logic

Decision States

pnpm check:imports

all boundaries clean

• ALLOW — All invariants passed, policy satisfied, approval valid (if required). Action authorized.

HOLD — Policy requires human approval that is absent or not yet submitted. Action queued.

• REJECT — One or more invariants failed. Action prohibited/blocked.

pnpm demo:web:validate

13 assertions pass

pnpm review:validate

110 assertions pass

2.2 Invariant Framework — Implemented Code

What it is: 12 unconditional enforcement rules that cannot be overridden by configuration or policy. Each invariant causes REJECT (or HOLD for INV-03) when its condition is not satisfied.

Source file: architecture/invariants/invariant-registry.yaml

Check command: pnpm check:invariants — confirms 12/12 are covered by tests

ID	Name	Linked Reason Code(s)	Test Coverage
INV-01	no_policy_pack_no_execution	NO_POLICY_PACK	execution-gate-service, adversarial-integrity
INV-02	no_provenance_no_action	NO_PROVENANCE	execution-gate-service, adversarial-integrity
INV-03	no_required_approval_no_release	REQUIRED_APPROVAL_MISSING, INVALID_APPROVAL_AUTHORITY, PRIVILEGED_AUTH_NOT_SATISFIED, APPROVAL_SIGNATURE_MISSING	execution-gate-service, misuse-scenarios, adversarial-integrity, contextual-boundary
INV-04	no_logging_no_execution	LOGGING_NOT_READY	execution-gate-service, misuse-scenarios, adversarial-integrity
INV-05	ai_non_authoritative	AI_CANNOT_AUTHORIZE	execution-gate-service, non-forgery, adversarial-integrity, contextual-boundary
INV-06	no_bypass_of_execution_gate	(none — structural)	non-forgery, misuse-scenarios
INV-07	immutable_decision_envelope	(none — artifact integrity)	execution-gate-service, adversarial-integrity
INV-08	stale_controls_block_sensitive_release	CONTROL_STALE_OR_INVALID	execution-gate-service, misuse-scenarios, adversarial-integrity
INV-09	trust_state_required	TRUST_STATE_INVALID	execution-gate-service, misuse-scenarios, adversarial-integrity
INV-10	prohibited_use_must_block	PROHIBITED_USE	execution-gate-service, misuse-scenarios, adversarial-integrity
INV-11	request_schema_and_action_class_valid	MALFORMED_REQUEST, UNKNOWN_ACTION_CLASS	execution-gate-service, misuse-scenarios, adversarial-integrity, fail-closed
INV-12	proposal_and_request_action_must_match	INVALID_PROPOSAL	execution-gate-service, misuse-scenarios, adversarial-integrity

What it is: Two JSON files that define the client's specific workflow, actors, approval chains, and action policies without requiring TypeScript changes.

File: deployment-starter/cerbaseal.config.json

```
{
  "authorityClasses": {
    "core": ["system", "ai", "analyst", "reviewer", "manager", "compliance_officer"],
    "extended": []
  },
  "workflowClasses": {
    "core": ["fraud_triage", "transaction_escalation", "account_hold_recommendation"],
    "extended": ["your_workflow_class"]
  },
  "actionClasses": {
    "core": ["allow", "hold", "reject", "escalate", "account_hold"],
    "extended": []
  }
}
```

File: deployment-starter/cerbaseal.policy.json

```
{
  "_schema": "cerbaseal-policy/v1",
  "actorMappings": {
    "Senior Fraud Analyst": "analyst",
    "Compliance Lead": "compliance_officer",
    "Operations Manager": "manager"
  },
  "approvalChains": {
    "your_workflow_class": ["analyst", "compliance_officer"]
  },
  "actionPolicies": {
    "your_workflow_class": {
      "escalate": "requires_approval",
      "account_hold": "requires_approval",
      "allow": "auto_allow",
      "hold": "auto_allow",
      "reject": "auto_allow"
    }
  }
}
```


Validation: Audit check 16 (pnpm audit:repo) parses policy JSON and validates structure

Status: Implemented — configuration is JSON, no TypeScript required for standard workflows

Governance pack templates (4 built-in):

- docs/client-adoption/templates/fraud-triage-template.md
- docs/client-adoption/templates/transaction-escalation-template.md
- docs/client-adoption/templates/account-hold-recommendation-template.md
- docs/client-adoption/templates/generic-human-approval-template.md

2.4 Audit Log — Implemented Code

What it is: An append-only JSONL log where every gate decision is recorded as a structured audit entry. Two backends: in-memory (development) and file-backed (production).

Item	Detail
Source path	src/services/audit/
File-backed implementation	examples/fraud-workflow-starter/ (demonstrates FileBackedAppendOnlyLogService)
Test file	test/persistent-audit-log.test.ts (12 tests)
Format	JSONL — one JSON object per line
Production log path (recommended)	/var/log/cerbaseal/workflow.jsonl
Status	Implemented and tested; file-backed path is production-suitable

2.5 Evidence Bundles — Implemented Code

What it is: An EvidenceBundle is produced with every gate decision. It contains the full request, the decision result, all checked invariants, reason codes, and a cryptographically chained audit entry reference.

Item	Detail
Source path	src/services/evidence/
Test file	test/audit-evidence-export.test.ts (6 tests)
Status	Implemented — evidence is first-class output, not logged separately

2.6 Proof Export and Verification — Implemented Code

What it is: A proof snapshot (proof-snapshot.json) that captures the full verification state of the repository at a point in time. The snapshot is cryptographically checksummed with two values: stableChecksum (enforcement state, stable across documentation changes) and manifestChecksum (full body including docs and examples).

Item	Detail
Export script	scripts/export-proof.ts
Verify script	scripts/verify-proof.ts
Export command	pnpm export:proof
Verify command	pnpm verify:proof
Output file	docs/reports/proof-snapshot.json
Current stableChecksum	e5aca8b2ad5f7abb528322be754d06463cb01367e038daf1472f925206c64e2c
Current manifestChecksum	53f5733305c6e585b21ed8c4239a1197fd48efb91da690044dc4f116def20cae
Status	Implemented — cryptographic verification confirmed

2.7 Replay / Verification — Implemented Code

Item	Detail
Source path	src/services/replay/
Snapshot test file	test/snapshots/enforcement-loop.snapshot.test.ts (41 tests)
Status	Implemented — snapshot tests confirm gate behavior is stable across changes

2.8 HTTP API and Review Portal — Implemented Code

What it is: A Node.js HTTP server that exposes the enforcement gate over HTTP and serves a browser-based review portal. All routes tested by review:validate (110 assertions).

Source file: examples/browser-demo/server.ts

Start command: pnpm demo:web

HTTP routes:

Route	Type	What It Does
GET /	HTML	Browser-based review portal (main dashboard)
GET /review	HTML	Review portal — partner/client-facing review view
GET /pilot	HTML	Pilot readiness page
GET /security	HTML	Security summary page
GET /deployment	HTML	Deployment information page
GET /one-page	HTML	One-page summary view
GET /api/reject	JSON	Execute REJECT scenario; returns full GateResult
GET /api/hold	JSON	Execute HOLD scenario; returns full GateResult
GET /api/allow	JSON	Execute ALLOW scenario; returns full GateResult
GET /api/review-summary	JSON	Structured review summary for portal
GET /api/pilot-readiness	JSON	Pilot readiness data object
GET /api/security-summary	JSON	Security summary data object
GET /robots.txt	text	Crawler control
GET /sitemap.xml	XML	Sitemap
GET /lms.txt	text	AI-readiness metadata
GET /assets/*	binary	Static assets

Test file: test/integration/review-portal-routes.test.ts (110 tests)

Validator: pnpm review:validate (110 assertions, all pass)

2.9 Deployment Starter — Implemented Code

What it is: A self-contained deployment package with configuration templates, a Dockerfile, Docker Compose file, and a verification script. Designed for client-controlled deployment.

Path: deployment-starter/

File	Purpose
deployment-starter/cerbaseal.config.json	Runtime configuration template (authorityClasses, workflowClasses, actionClasses)
deployment-starter/cerbaseal.policy.json	Policy template (actorMappings, approvalChains, actionPolicies)
deployment-starter/Dockerfile	Container image definition
deployment-starter/docker-compose.yml	Docker Compose orchestration file
deployment-starter/index.ts	Entry point for the deployment server
deployment-starter/verify.ts	Verification script — runs REJECT, HOLD, ALLOW scenarios programmatically
deployment-starter/README.md	Deployment-starter specific README

Verification command: tsx deployment-starter/verify.ts

Status: Implemented — Docker Compose and Node.js Direct paths both supported

2.10 Setup Wizard — Implemented Code

What it is: An interactive CLI wizard that walks a partner or client through configuration, writing cerbaseal.config.json and cerbaseal.policy.json interactively.

Item	Detail
Source	scripts/setup.ts
Command	pnpm setup
Status	Implemented

2.11 Partner Kit Documentation — Documentation

Path: docs/partner-kit/

File	Audience	What It Covers
00-OVERVIEW.md	All partners	Kit index, guided independence model, certification levels, tools
01-sales-brief.md	Sales staff	Positioning, qualification (3 questions), buyer persona map
02-technical-brief.md	Technical staff	Architecture, invariants, evidence chain
03-deployment-guide.md	Technical lead	Step-by-step deployment, verification, deployment checklist (Appendix)
04-pilot-guide.md	Partner lead	1-day structured pilot delivery, commercial framing, session guide
05-support-guide.md	Partner support	10 common issues with diagnosis and resolution
06-objection-handling.md	Sales/technical	8 enterprise objections with persona-calibrated responses
07-certification-framework.md	All partners	Three levels (Deploy, Configure, Lead Pilots), practical exercises
08-pricing-and-commercial-model.md	Sales leads, principals	5 pilot tiers, annual licensing, channel economics, negotiation posture
09-market-positioning.md	Sales leads, principals	Category placement, comparable benchmarks, regulatory context, source notes
10-adoption-roadmap.md	Lamont Labs, partners	Build priorities, founder independence metrics, next-steps timeline
11-risk-register.md	All	7 risk areas, neutral external language, out-of-scope list, communication posture
12-partner-discussion-brief.md	Partner principals	Pre-call Q&A, talking points, negotiation posture for commercial discussions

Status: All 13 documents are implemented and complete.

2.12 Client Adoption Documentation — Documentation

Path: docs/client-adoption/

File / Directory	What It Covers
workflow-mapping-workbook.md	Sections A–M for facilitated workflow mapping sessions
policy-pack-authoring-guide.md	Step-by-step policy JSON authoring
quickstart-deployment-guide.md	3 time-boxed deployment paths
pilot-delivery-playbook.md	Week-by-week pilot delivery guide (Tier 2 = 5 weeks, target "d8 Jesse hours")
pilot-sizing-and-pricing-framework.md	Internal pricing tiers (EUR), pricing factors, recommended first offer range
pilot-success-framework.md	Success and failure criteria definition
client-discovery-script.md	Full script for discovery/qualification calls
client-qualification-scorecard.md	Green/Yellow/Red readiness scoring
client-readiness-assessment.md	Structured readiness assessment
client-admin-guide.md	Client administrative responsibilities
onboarding-sequence.md	Onboarding phases and sequence
support-boundaries.md	What is self-service vs. escalation
founder-dependency-reduction-plan.md	8-layer dependency map with mitigation status
eu-ai-act-nis2-mapping-support.md	Regulatory mapping support
self-service-configuration-wizard-spec.md	Specification for future policy wizard (roadmap)
frequently-asked-objections.md	Extended objection handling reference
troubleshooting-guide.md	Troubleshooting beyond the support guide
train-the-trainer-program.md	Program for training partner delivery staff

templates/	4 governance pack templates (fraud triage, transaction escalation, account hold, generic approval)
training/	8 training materials (exec overview, onboarding agenda, operator guide, reviewer guide, admin guide, FAQ, common errors, getting started)

2.13 Diagnostic Tools — Implemented Code

Tool	Command	What It Does	Status
Diagnostic report service	src/services/diagnostics/	Generates structured diagnostic report	Implemented
Diagnostic report test	test/diagnostic-report-service.test.ts (5 tests)	Tests report output	Passing
Repo audit	pnpm audit:repo	16 automated governance checks	Implemented, 16/16 pass
Invariant coverage check	pnpm check:invariants	Verifies all 12 invariants are covered	Implemented, 12/12 pass
Import boundary check	pnpm check:imports	Verifies no architectural boundary violations	Implemented, clean
Evidence report generator	pnpm generate:evidence-report	Human-readable compliance evidence report	Implemented
Support readiness validator	pnpm demo:support:validate	13 assertions against support readiness	Implemented, 13/13 pass

2.14 Review Portal — Implemented Code

What it is: A browser-based portal served at GET / with interactive ALLOW/HOLD/REJECT scenario views, review summary, pilot readiness data, and security summary. Tested with 110 assertions.

Item	Detail
Server source	examples/browser-demo/server.ts
Review portal data	src/domain/review-portal-data.ts
Review portal data source	examples/browser-demo/review-portal.ts — exports REVIEW_SUMMARY, PILOT_READINESS, SECURITY_SUMMARY
Test file	test/integration/review-portal-routes.test.ts (110 tests)
Validator	pnpm review:validate (110 assertions)
Start command	pnpm demo:web
Status	Implemented, 110 assertions pass

2.15 Starter Kit Library — Implemented Code

Path: examples/

Reference: examples/INTEGRATION-GUIDE.md

Starter Kit	Path	What It Demonstrates	Estimated Integration Time
Express middleware	examples/express-middleware/	cerbaSealGate() middleware; ALLOW !' next(), REJECT !' 403, HOLD !' 202	30 minutes
Webhook adapter	examples/webhook-adapter/	HTTP receiver; POST /event !' gate !' POST callback; async approval loop	1 hour
Async queue	examples/async-queue/	In-memory JobQueue; enqueue !' HOLD !' approve !' ALLOW	1–2 hours
Audit export	examples/audit-export/	Read JSONL log; verify chain integrity; export counts by event type	30 minutes
REST API starter	examples/rest-api-starter/	Standalone HTTP gate server; evaluate endpoint; decision history	1 hour
Financial approval starter	examples/financial-approval-starter/	HOLD !' manager approves !' ALLOW; ApprovalArtifact construction	30 minutes
Agent integration starter	examples/agent-integration-starter/	AI self-authorization prevention (correct vs wrong patterns)	30 minutes

What Docker Config Pull Without Code Changes

Status: Implemented — build-ready, not independently production-certified

Starter Kit			
Item	Detail		
Files	deployment-starter/Dockerfile, deployment-starter/docker-compose.yml		
Prerequisites	Docker Desktop or Docker Engine, Docker Compose v2+		
Start command	docker compose up (from deployment-starter/)		
Configuration	Edit cerbaseal.config.json and cerbaseal.policy.json before starting		
Verification	tsx deployment-starter/verify.ts — runs 9 assertions (REJECT, HOLD, ALLOW scenarios)		
Production considerations	Network access control, persistent volume mapping for audit log, restart policy — all operator responsibility		
Known gaps	Network-level authentication of API callers is operator responsibility; no built-in rate limiting or WAF		
Core workflow classes (built-in, can be supplemented)	examples/consumer-integration/parsa	transaction_escalation, account_hold, recommendation	Implemented
Core action classes (built-in, can be supplemented)	examples/auditor-view/	Auditor-facing view of audit log	Not specified
Support readiness	examples/support-readiness/	Support readiness demonstration	Not specified

In cerbaseal.policy.json
Status: Implemented — pilot-ready, not independently production-certified
Integration test file: test/integration/external-signal-examnples.test.ts (16 tests)

Item	Detail		
Prerequisites	Node.js 18+ (tested on v22.22.0), pnpm		
Install command	git clone <repo> && cd cerbaseal-core && pnpm install		
Verify install	pnpm test (432/432 must pass), pnpm audit:repo (16/16 must pass)		
Start command	pnpm demo:web (PORT env variable configures port, default 3001)		
Config setup	pnpm setup (interactive wizard) or manually edit JSON files		
Verification	tsx deployment-starter/verify.ts		
Deployment guide	docs/partner-kit/03-deployment-guide.md, docs/client-adoption/quickstart-deployment-guide.md		
Known gaps	Windows deployment not tested; Node.js version must be 18.0+		
Commercial framework PDF	pnpm generate:commercial-pdf	Commercial framework PDF	Implemented

Capability	Why Code Is Required
Adding a new enforcement invariant	INV-01 through INV-12 are unconditional TypeScript enforcement rules; a new one requires a new TypeScript condition and new tests
Status: Implemented	
Adding a new ProposalSourceKind	Only "ai" and "deterministic_rule" are currently valid; new values require TypeScript source changes

Item	Detail
Path	examples/rest-api-starter/
What it provides	Standalone HTTP gate server with evaluate endpoint and decision history
Estimated setup	1 hour
Adding new API routes to the review portal	examples/browser-demo-server/vars requires TypeScript changes

4.4 Local Demo / Review Portal	
New governance pack templates (as live templates)	Adding to the core arrays in config is a TypeScript change; extended arrays handle client additions
Status: Implemented — demo-ready	

Item	Detail
Command	pnpm demo:web
Serves	Browser-based review portal at localhost (PORT env var or 3001)
Routes	/, /review, /pilot, /security, /deployment, /one-page, /api/reject, /api/hold, /api/allow, /api/review-summary, /api/pilot-readiness, /api/security-summary
Validation	pnpm review:validate (110 assertions)
Purpose	Demonstrates enforcement gate for client-facing review; not a production entry point

4.5 Client-Controlled Deployment Posture

Status: Documented — this is the recommended production architecture

The deployment model is client-controlled: the client provides and owns the hosting environment, network layer, identity provider, and persistence volume. CerbaSeal runs as a service within the client's perimeter. No data leaves the client's environment.

Supporting docs:

- docs/partner-kit/03-deployment-guide.md — step-by-step for partner-led deployment

Reference document: docs/client-adoption/pilot-delivery-playbook.md

Format: Tier 2 Controlled Workflow Pilot = 5 weeks. Adjust to 3 weeks for Tier 1, 8 weeks for Tier 3.

Jesse hours target: "d 8 total across full Tier 2 pilot.

- docs/client-adoption/quickstart-deployment-guide.md — 3 time-boxed deployment paths
- docs/client-adoption/client-admin-guide.md — client's operational responsibilities

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Pre-Pilot — Agreement and Readiness

4.6 Deployment Decision Checklist (from Appendix B, `docs/partner-kit/03-deployment-guide.md`)

Activity	Client	Partner (Line Axia / certified partner)	Lamont Labs
Complete readiness assessment	Score on qualification scorecard	Use client-qualification-scorecard.md	Confirm technical readiness
Sign working agreement	Sign	Prepare and present	Jesse signs; defines support hour allocation
Agree success criteria	Sign	Draft from pilot-success-framework.md	Review
Confirm named contacts	Provide technical owner	Provide account contact	Confirm Jesse support window
Enforcement gate logic (evaluate())	Production-grade logic	432 tests including adversarial, 12/12 invariants, snapshot tests	Deliverables: Signed working agreement, completed readiness assessment, named contacts, agreed success criteria document
Audit log (file-backed)	Production-suitable	12 tests; standard JSONL append-only pattern	
Docker Compose deployment	Pilot-ready	Operator adds network controls, monitoring, restart policy	Supporting files: docs/client-adoption/client-qualification-scorecard.md, docs/client-adoption/client-readiness-assessment.md, docs/client-adoption/pilot-success-framework.md
Third-party security review	Future roadmap (v0.2.0)	Not yet completed	
HTTP API (review portal server)	Demo / pilot	Not a hardened production server; no rate limiting, no auth on API callers	Phase 1 — Discovery (Week 1, Days 1–2)
PKI-backed signing	Future roadmap (v0.2.0)	Presence-checking only in v0.1.0	
Activity	Duration		Owner
30-minute kickoff session (onboarding agenda)	30 min		Partner facilitates; Jesse not required
Browser demo (pnpm demo:web)	During kickoff		Partner demonstrates
Distribute role-specific training materials	At kickoff		Partner
Capture early questions in writing	During kickoff		Partner

Supporting files: docs/client-adoption/training/30-minute-onboarding-agenda.md, docs/client-adoption/training/ (8 training materials)

Jesse hours: 0 (unless client specifically requests)

Phase 2 — Workflow Mapping (Week 1, Days 3–4)

Activity	Duration	Owner
Workflow mapping session (sections A–M of workbook)	2–3 hours	Partner facilitates; client operational lead + technical owner
Complete CerbaSeal Field Map	During session	Partner + client
Resolve ALLOW / HOLD / REJECT conditions	During session	Client decision; partner structures

Output: Completed policy pack worksheet mapping all four sections of cerbaseal.policy.json

Supporting files: docs/client-adoption/workflow-mapping-workbook.md

Jesse hours: Optional standby — answers technical questions asynchronously

Phase 3 — Configuration (Week 1–2)

Activity	Duration	Owner
Policy pack authoring	1–3 hours	Partner; uses policy-pack-authoring-guide.md
pnpm audit:repo validation (check 16)	Minutes	Partner
Client review of policy	30 min	Client reviews actor names and approval chains

Supporting files: docs/client-adoption/policy-pack-authoring-guide.md, deployment-starter/cerbaseal.policy.json (template)

Governance templates: 4 built-in templates in docs/client-adoption/templates/

Jesse hours: 0 for standard workflows; possible async review

Phase 4 — Deployment (Week 2)

Activity	Duration	Owner
Clone and install	15 min	Partner engineer
pnpm test pass	Minutes	Partner engineer
pnpm audit:repo pass	Minutes	Partner engineer
Docker Compose or Node.js Direct setup	30–60 min	Partner engineer
Audit log path configuration	5 min	Partner + client confirms path
tsx deployment-starter/verify.ts (9 assertions)	Minutes	Partner

Supporting files: docs/partner-kit/03-deployment-guide.md, docs/client-adoption/quickstart-deployment-guide.md

Jesse hours: 0 for standard deployment; 1–2 for non-standard environments

Phase 5 — Validation (Week 2–3)

Activity	Duration	Owner
ALLOW scenario verification	15 min	Partner + client
HOLD scenario verification (with approver)	30 min	Partner + client + approver representative
REJECT scenario verification	15 min	Partner + client
GET /health confirms auditChainValid: true	5 min	Partner
Representative pilot requests	1–2 hours	Client + partner

Supporting files: docs/partner-kit/04-pilot-guide.md (Session 3 + 4 detail), deployment-starter/verify.ts

Phase 6 — Pilot Operation (Week 3–4)

Activity	Duration	Owner
Client operates workflow through gate	Ongoing	Client
Partner provides Tier 1/2 support	As needed	Partner
Issues logged and triaged	As needed	Partner
Tier 3 escalation (if needed)	As needed	Lamont Labs

Supporting files: docs/partner-kit/05-support-guide.md, docs/client-adoption/support-boundaries.md

Jesse hours: 0–2 (Tier 3 escalations only)

Phase 7 — Evidence Closeout (Week 5)

Activity	Duration	Owner
pnpm export:proof	Minutes	Partner
pnpm verify:proof (checksums confirmed)	Minutes	Partner
pnpm generate:evidence-report	Minutes	Partner
Evidence review session with compliance stakeholders	60 min	Client + partner
Document issues, tickets, next-step recommendation	30 min	Partner + Lamont Labs
Sign pilot closeout summary	Client + partner	Both

Deliverables: proof-snapshot.json, JSONL audit log, evidence report, pilot closeout summary

Jesse hours: 1–2 (evidence review participation, if requested)

Training Support Materials

Reference document: docs/client-adoption/founder-dependency-reduction-plan.md

Source: Pilot delivery playbook (Tier 2 = 5-week pilot, target "d 8 Jesse hours total)

Material	Audience	Format	Path
10-minute executive overview	CRO, CISO, executives	Self-directed reading	training/10-minute-executive-overview.md
Scenario	Total Jesse Hours	Basis	
Best case	0–2 hours	Standard workflow, strong technical partner, no Tier 3 issues	
Realistic (target)	4–8 hours	One async technical review, one evidence session, one Tier 3 query	
Difficult case	12–20 hours	Non-standard environment, multiple Tier 3 escalations, new invariant discussion	
FAQ	All	Self-directed	training/fag.md
The playbook target is "d 8 hours. If actual hours exceed 8 for a Tier 2 pilot, the playbook calls for scope and support model review.	Technical / operations	Self-directed	training/common-errors-and-fixes.md
Common errors and fixes			
Getting started guide	Technical	Self-directed	training/getting-started-guide.md

Jesse Involvement Breakdown by Phase

Section 6 — Founder Dependency Analysis			
Phase	Activity	Jesse Required?	Replaceable By
Pre-pilot	Sales qualification	No	client-discovery-script.md, client-qualification-scorecard.md
Pre-pilot	Agreement signing	Yes (signs working agreement)	Cannot be delegated — Jesse is the legal counterparty
Pre-pilot	Technical readiness confirmation	Async only	Documentation-first; npm audit:repo self-checks
Week 1	30-minute kickoff	No (unless client requests)	training/30-minute-onboarding-agenda.md
Week 1	Workflow mapping	No (optional standby)	workflow-mapping-workbook.md sections A–M
Week 2	Policy authoring	No	policy-pack-authoring-guide.md, 4 templates
Week 2	Deployment	No (except non-standard environments)	quickstart-deployment-guide.md, 03-deployment-guide.md
Week 2–3	Verification	No	verify.ts script, audit check 16
Week 3–4	Pilot support (Tier 1/2)	No	Partner uses 05-support-guide.md
Week 3–4	Tier 3 escalation	Yes — irreplaceable	Cannot be delegated for core invariant or source changes
Week 5	Evidence review	Optional (1–2 hrs)	Partner can lead using evidence report and templates
Week 5	Closeout summary	No	Partner produces; Jesse reviews async

Top 10 Remaining Founder-Dependency Points

#	Dependency	Reduction Path	Current Status
1	Legal counterparty on working agreement	Partner agreement framework with IP protections	Not yet templated
2	Tier 3 escalation for invariant questions	Extended technical documentation, FAQ expansion	Partially mitigated by 02-technical-brief.md
3	New workflow JSON authoring for complex cases	Policy pack builder (Priority 1 build item)	Roadmap only — spec exists at self-service-configuration-wizard-spec.md
4	First 1–2 workflow mapping sessions for new partners	Train-the-trainer program	' Done — train-the-trainer-program.md
5	Evidence review explanation for non-technical stakeholders	Replay and evidence explorer (Priority 4 build item)	Roadmap only
6	Non-standard deployment environments	Extended troubleshooting guide	' Done — troubleshooting-guide.md
7	Technical validation request from sophisticated clients	Expanded technical brief and Q&A	Partially mitigated by 02-technical-brief.md
8	Security review questions during procurement	Third-party security review package (Priority 6 build item)	Roadmap — not yet completed
9	First paid pilot commercial scoping	Pricing framework, partner discussion brief	' Done — 08-pricing-and-commercial-model.md, 12-partner-discussion-brief.md
10	Pilot closeout and evidence presentation	Evidence closeout template (roadmap)	Partially mitigated by generate:evidence-report

Supporting files for this section:

- docs/client-adoption/founder-dependency-reduction-plan.md — 8-layer dependency map with status for each
- docs/client-adoption/pilot-delivery-playbook.md — explicit Jesse hours per week
- docs/client-adoption/train-the-trainer-program.md

Recommended First Pilot Shape

File	Audience	Purpose	Completeness
docs/partner-kit/00-OVERVIEW.md	All	Kit index, guided independence model, tool reference	Complete
Element	Specification		
Workflow scope	One workflow, one decision path, one approval model		
Duration	45–90 days (5-week delivery playbook)		
Deployment mode	Docker Compose or Node.js Direct, client-controlled environment		
Decision states validated	ALLOW, HOLD (with live approver), REJECT		
Evidence output	Proof snapshot (proof-snapshot.json), JSONL audit log, evidence report		
Support model	Partner-led Tier 1/2; Lamont Labs Tier 3 for core issues only		
Jesse hours target	"d 8 hours across full pilot		
handling.md	technical)		
docs/partner-kit/01-INTRO.md	Supported Pilot Workflow Archetypes (Built-in)		Complete
framework.md	Level 1 (Deploy), Level 2 (Configure), Level 3 (Lead Pilots) with exercises		
docs/partner-kit/08-pricing-and-sales-leads.md	Sales leads	5 pilot tiers, annual licensing, 4 channel models, pilot package	Complete
Workflow	Built-in Class Name	Starter Kit	
Fraud triage	fraud_triage	examples/fraud-workflow-starter/	
Transaction escalation	transaction_escalation	examples/financial-approval-starter/	
Account hold recommendation	account_hold_recommendation	examples/financial-approval-starter/	
AI agent tool execution	(custom via extended)	examples/agent-gate/, examples/agent-integration-starter/	
Generic approval workflow	(custom via extended)	examples/rest-api-starter/	
brief.md			

Pilot Success Criteria
Certification Levels

Reference: docs/client-adoption/pilot-success-framework.md

Level	Name	What It Proves	Minimum Before
Level 1	One defined workflow mapped to CerbaSeal request, config, and policy structures	Can deploy a policy package for a new client workflow without assistance	Any client deployment
Level 2	ALLOW, HOLD, and REJECT Comigore	Can add a policy package for a new client workflow without assistance	Unsupervised pilot delivery
Level 3	Proof-snapshot.json exported with checksums verified	Can run a full pilot from kickoff to evidence closeout with appropriate escalation	Leading engagement solo

- Reference: docs/partner-kit/07-certification-framework.md
- Evidence package reviewed with operational and compliance stakeholders
 - Operational responsibilities confirmed and documented (monitoring, escalation path, admin)
 - Issues and support tickets documented with disposition

Pilot Failure Criteria (Out of Scope Declared)
Section 8 — Pilot Readiness

If any of the following are expected, the pilot scope must be explicitly renegotiated before work begins:

- Production certification or formal regulatory compliance certification
- Unlimited custom development
- 24/7 support without written scope
- Multi-workflow scope in a Tier 2 engagement without Tier 4 pricing
- AI model quality, correctness, or bias validation
- Built-in network-level API authentication (operator responsibility)

Evidence Outputs

Output	Command	What It Contains
Proof snapshot	pnpm export:proof	stableChecksum, manifestChecksum, test results, invariant coverage, validator results, git commit
JSONL audit log	Written during operation	One entry per gate decision; SHA-256 chained
Evidence report	pnpm generate:evidence-report	Human-readable compliance evidence
Verification confirmation	pnpm verify:proof	Confirms checksums match; confirms snapshot has not been tampered with

Escalation Path

Important: This section contains only internal facts. No prices are recommended here. Pricing implications are noted neutrally.

Implementation Effort Facts

Tier	Who	What It Covers
Fact	Evidence Source	
Standard integration time (express middleware or REST API starter)	30–60 minutes (per examples/INTEGRATION-GUIDE.md)	
Webhook adapter integration	~1 hour (per examples/INTEGRATION-GUIDE.md)	
Async queue integration	1–2 hours (per examples/INTEGRATION-GUIDE.md)	
Policy pack authoring (standard workflow)	Covered in workflow mapping session (2–3 hours total for mapping + authoring)	
Deployment setup (Docker Compose or Node.js Direct)	30–60 minutes for standard environment	
Full Tier 2 pilot delivery (5 weeks with experienced partner)	Partner and client time; Jesse "d 8 hours	
Services-to-software ratio	Pilot is services-heavy (discovery, mapping, deployment, validation, training, evidence, support); annual license is software-light	

Support Effort Facts

Fact	Evidence Source
Tier 1/2 issues solvable without Jesse	80% target (per docs/partner-kit/00-OVERVIEW.md)
10 most common partner support issues documented	docs/partner-kit/05-support-guide.md
Support readiness validator	pnpm demo:support:validate — 13 assertions pass
Escalation discipline	Tier 3 = new invariants, unresolved defects, source-level changes — reserved
Support boundaries document	docs/client-adoption/support-boundaries.md — explicitly defines what is self-service vs. escalation

Deployment Complexity Facts

Fact	Evidence Source
Two supported deployment paths	Docker Compose and Node.js Direct (deployment-starter/)
Deployment verification script	tsx deployment-starter/verify.ts — 9 assertions (REJECT, HOLD, ALLOW)
Deployment checklist	13 steps with owner assignments (docs/partner-kit/03-deployment-guide.md Appendix)
Client owns network perimeter	No built-in network-level API caller authentication — operator responsibility
Persistent audit log	File-backed JSONL — client provides persistent volume
Windows deployment	Not tested in v0.1.0
Non-standard environment risk	Additional Jesse time may be required

Configuration Complexity Facts

Fact	Evidence Source
Configuration is JSON — no TypeScript required for standard workflows	cerbaseal.config.json, cerbaseal.policy.json
4 built-in governance pack templates	docs/client-adoption/templates/
Policy validation built into audit check 16	pnpm audit:repo check 16
Policy authoring guide exists	docs/client-adoption/policy-pack-authoring-guide.md
Policy pack builder is roadmap (not yet built)	docs/client-adoption/self-service-configuration-wizard-spec.md (spec only)
Configuration errors are a common support issue	docs/partner-kit/05-support-guide.md — 10 common issues

Evidence Output Value Facts

No final prices are included. This table provides the inputs that should inform pricing decisions.

Offer Component	Client Receives	Work Required	Who Performs	Reusable or Custom	One-time or Recurring	Support Burden	Evidence Value	Pricing Implication
Discovery / Readiness Assessment	Fit assessment, workflow candidate, deployment path recommendation, pilot scope	1×60-min discovery call, readiness scoring, summary report	Partner facilitates; Lamont Labs reviews async	Reusable process, custom output	One-time	Very low	Low	Should cover partner discovery hours and preparation
1-Day Express Pilot	Governed workflow, ALLOW/HOLD/REJECT verified, proof snapshot	5 sessions (workflow mapping, policy, deployment, integration, verification)	Partner leads, client technical owner participates	Template reusable; workflow policy is custom	One-time	Low during, low after	High (exportable evidence)	Must cover 1 partner day + preparation + post-day closeout
Controlled Workflow Pilot (Tier 2)	One production-adjacent workflow, guided deployment, evidence package, bounded support window	5-week delivery playbook; all onboarding phases	Partner leads all phases; Jesse "d 8 hours	Playbook reusable; all client artifacts custom	One-time	Moderate (Tier 1/2 partner; Tier 3 reserved)	High (audit log, proof snapshot, evidence report)	Must cover full 5-week partner engagement + Jesse time + materials
Deployment Support	Deployment path setup, config/policy validation, health check verification	30–90 min for standard environment; variable for non-standard	Partner engineer	Mostly reusable	One-time	Low after successful deployment	Moderate	Should reflect actual deployment hours; premiums for non-standard environments
Configuration Support	Policy pack authoring for client workflow; actor mapping; approval chains	2–4 hours for standard workflow	Partner (post-Level 2 cert)	Template reusable; workflow mapping is custom	One-time	Low after authoring; higher if complex policy	Moderate	Reflects policy complexity and mapping session hours
Annual License	Software access, version updates, usage rights	Lamont Labs maintains source, releases updates	Lamont Labs	Fully reusable	Recurring	Depends on support tier	High (continuing enforcement value)	Should reflect enforcement value, not pilot services; separate from one-time implementation
Partner License / Channel Fee	Partner right to resell, deploy, and deliver CerbaSeal	Lamont Labs provides certification path, kit updates, Tier 3 backstop	Lamont Labs	Reusable	Recurring (annual)	Low if partner is certified; higher for uncertified partners	High if partner can deliver repeatedly	Should reflect certification investment and Tier 3 support access
Additional Workflow	Second governed workflow added after first pilot	New mapping session, policy extension, verification	Partner	Template reusable; workflow policy is custom	One-time (per workflow)	Lower than first workflow (no deployment work)	Additional evidence value	Should be meaningfully less than first workflow (no deployment overhead)
Support Package	Bounded support hours per month / quarter	Triage, diagnosis, Tier 2 resolution	Partner Tier 1/2; Lamont Labs Tier 3 (rarely)	Mostly reusable	Recurring	Moderate (depends on client technical maturity)	Moderate (audit trails maintained)	Should be based on observed support hours per pilot + buffer
Evidence / Audit Package	Periodic proof snapshot, evidence report, audit log export, review session	export:proof, generate:evidence-report, review facilitation	Partner facilitates; client stakeholders attend	Commands reusable; content is custom	One-time or periodic	Low (commands are scripted)	High (compliance value)	Should reflect time for review session facilitation, not command execution
Security / Procurement Package	Security summary, known-limitations disclosure, future security review context	Compile security summary, SECURITY_SUMMARY from review portal	Partner compiles; Jesse reviews	Mostly reusable	One-time per procurement cycle	Low	High (accelerates enterprise procurement)	Should be priced to recover preparation time; unlocks larger engagements
Custom Integration	Client-specific integration beyond starter kits	TypeScript integration development, testing	Lamont Labs or partner engineer (if Level 2+ certified)	Custom	One-time	Moderate to high (ongoing maintenance risk)	Varies	Time-and-materials; not bundled in standard pilot
Roadmap / Custom Development	New invariant, new ProposalSourceKind, new API capability	TypeScript source changes + new tests	Lamont Labs only	Not reusable by default	One-time	High (Jesse involvement required)	High if commercially relevant	Time-and-materials; requires written scope; IP treatment must be agreed in advance

Prices shown below are intended only for the many steps in the internal reference notes only (per docs/partner-kit/09-market-positioning.md). Prices marked [VERIFY] require external lookup.

Security			
Source	What to Look For	URL or Search Path	Repo Notes
Item	Status	External Description	Roadmap
Third-party security review	Not yet completed	Formal third-party security review is planned as a future milestone	v0.2.0
PKI-backed cryptographic signing of DecisionEnvelope	Current scope: presence-checking only	immutableSignature is checked for presence and non-emptiness; cryptographic key validation is a roadmap item	v0.2.0
HTTP API caller authentication	Operator responsibility	Network-level access control is implemented by the operator using gateway, VPN, or private network	Operator responsibility
Rate limiting, circuit breaking, WAF	Operator responsibility	Denial-of-service resilience is implemented at the deployment perimeter by the operator	Operator responsibility
Key management infrastructure	Not built	Approval signature key management is an operator and future platform concern	Future roadmap
Marketplace	usage basis	marketplace/pp/prodview-x67krdatcdday	on contract duration and usage
Integration			
Fiddler AI — Control Plane	AI control-plane positioning language; pricing if listed	fiddler.ai/control-plane	Repo notes: "Enforceable policy, auditable governance, centralized oversight" language
Item	Status	External Description	
Identity provider integration	Boundary responsibility	CerbaSeal evaluates TrustState but does not operate the client IdP; integration boundary and required identity assertions are defined during workflow mapping	
Production monitoring integration	Operator responsibility	Operational monitoring is environment-specific; CerbaSeal provides a health endpoint (GET /health) and audit chain verification	
Windows deployment	Not tested	Windows deployment is outside current tested scope; support posture is unclear	
AI governance platform pricing (general)	Credo, Monitaur, Holistic AI, other AI governance tools	G2 "AI Governance" category; AWS Marketplace	[VERIFY] Use for category validation
Configuration and Tooling			
Policy enforcement	OPA/Styra ecosystem; Permit.io;	AWS Marketplace; vendor	[VERIFY] Styra is confirmed; others require
Item	Status	External Description	
Policy pack builder (wizard)	Roadmap	A guided policy configuration experience is a Priority 1 build item; current approach is JSON with partner guidance and validation	
Workflow mapping wizard	Roadmap	Turns discovery into a repeatable automated process; current approach uses the workflow mapping workbook	
Governance pack library (expanded)	Roadmap	Additional pre-built templates for finance, insurance, KYC, claims, and AI agent workflows	
Replay and evidence explorer	Roadmap	A visual evidence explorer for non-engineering stakeholders; current approach is generate:evidence-report	
Partner deployment automation	Roadmap	Further reduces setup time; current approach uses documented deployment paths and verify.ts	
Contract values are often not publicly available. The pricing framework should be used as a structured discussion model and refined after direct partner feedback, first pilot scoping, and observed support effort.			
Commercial and Operations			
Item	Status	External Description	
Founder dependency in early pilots	Current scope	Tier 3 support and roadmap changes require Lamont Labs involvement; guided independence model reduces routine dependency	
Pricing uncertainty	Current scope	Private enterprise pricing is not fully public across comparable vendors; pricing framework should be revised after first pilots and observed support effort	
Annual support SLA	Not yet defined	Formal annual support SLA will be defined before first annual license agreement	
Partner agreement framework	Not yet templated	IP ownership, territory, and channel terms require written agreements; IP stays with Lamont Labs	

Section 13 — Final Verification Commands

[illegible]

Check	Command	Result	Notes
Test suite	pnpm test	432/432 PASS, 17 files	~5.2 seconds
Repo audit	pnpm audit:repo	16/16 PASS	All checks clean
Invariant coverage	pnpm check:invariants	12/12 covered	All INV-01–INV-12
Proof export	pnpm export:proof	PASS	Writes docs/reports/proof-snapshot.json
Proof verify	pnpm verify:proof	VERIFIED	Checksums match
Demo web validator	pnpm demo:web:validate	106 assertions PASS	
Demo support validator	pnpm demo:support:validate	13 assertions PASS	
Review portal validator	pnpm review:validate	110 assertions PASS	
Import boundary check	pnpm check:imports	PASS	All 29 src files clean
TypeScript compile	tsc --noEmit	PASS	No errors

Stable checksum (enforcement state): e5aca8b2ad5f7abb528322be754d06463cb01367e038daf1472f925206c64e2c

Manifest checksum (full body): 53f5733305c6e585b21ed8c4239a1197fd48efb91da690044dc4f116def20cae

Commit: 485cde3faaa0524ffd5f5b6132673789f89adab9

Branch: main

Stale Documentation Found

None. Audit check 14 (No stale test-count references in documentation) passed — all documentation references match the actual test count of 432.

Failed Commands

None. All commands above produced expected output.

Pricing-Relevant Uncertainties Requiring External Research

- Arthur AI, Credo AI current pricing — Both use private-offer models on AWS Marketplace. Prices are not publicly listed. Must be obtained from analyst sources (G2, Vendr, Gartner) or direct vendor inquiry.
- Vanta, OneTrust, Securix same pricing — Compliance automation category relevant. Relevant for buyer-budget calibration. Not in repo. Requires external research.
 - Current Synapse AI pricing — Repo cites \$5,000/month as of research date. Verify against current listing at [aws.amazon.com/marketplace/pp/prodview-vr5fms3c6y4c](#).
 - Current Workday pricing — Repo cites \$13,750/year. Verify against current listing at [aws.amazon.com/marketplace/pp/prodview-vr5fms3c6y4c](#).
 - Enterprise proof of copying norms (45–90 day) — No repo evidence for what regulated enterprises typically pay for a governed PoC. Forrester or Gartner research recommended.
 - Current partner services margin benchmarks — Repo cites 25%–45% professional-services margins from an AWS/Forrester source. Verify currency.
 - Annual license pricing norms for AI governance platforms — Credo AI and Arthur AI use private offers; public benchmarks are unavailable. Analyst reports or aggregator data (Vendr) recommended.